

EC1208

Principles of Macro-Economy

Summer 2025 Examination

Paper 1 · 1.5 hours · Section A: answer 2 of 3 (40%) · Section B: answer 1 of 2 (60%)

Module	EC1208 — Principles of Macro-Economy
Session	Summer 2025
Structure	Section A: three short questions (100 marks each) — answer any two, together worth 40% of the total. Section B: two long questions, each with four 25-mark parts — answer any one, worth 60% of the total.
This scheme covers	All three Section A questions AND both Section B questions in full, so you can study every possible combination rather than just the ones you'd pick on the day.

A note on Question A1. Its three sub-parts (i, ii, iii) aren't individually mark-weighted on the paper — only the total of 100 marks is given. This scheme allocates 25 / 30 / 45 marks across (i), (ii) and (iii) respectively, reflecting that part (iii) asks for the deepest theoretical explanation; check with your lecturer for the department's own exact split if one exists.

Section A — Short Questions

Answer 2 of 3 · 40% of total marks · each question worth 100 marks (20% of the paper)

Question A1

Aggregate Production Function (100 Marks)

The following table shows the change in GDP in Lithasia with changes in efficiency units of human capital.

GDP (in millions of dollars)	Stock of physical capital (units)	Efficiency units of labor
100	15,000	16,000
150	15,000	20,000
180	15,000	24,000
200	15,000	28,000
210	15,000	32,000

A1 (i)

25 marks

(i) Comment on the rate of change in GDP as the economy uses more efficiency units of labor.

FULL MODEL ANSWER

The Data

Change in efficiency units of labor (H)	Change in GDP
16,000 → 20,000 (+4,000)	100 → 150 (+50)
20,000 → 24,000 (+4,000)	150 → 180 (+30)
24,000 → 28,000 (+4,000)	180 → 200 (+20)
28,000 → 32,000 (+4,000)	200 → 210 (+10)

Physical capital (K) is held constant throughout at 15,000 units, so this table isolates the effect of labor alone. Each row adds exactly the same amount of efficiency units of labor (+4,000), yet the resulting increase in GDP steadily shrinks: +50, then +30, then +20, then only +10. GDP is still rising overall — more efficiency units of labor always raises GDP, consistent with the “more is better” property of the aggregate production function — but each additional block of labor adds progressively less to output than the last.

This is the Law of Diminishing Marginal Product at work. With physical capital fixed at 15,000 units, each additional efficiency unit of labor has progressively less capital to work with on average — there are more workers (in efficiency terms) sharing the same fixed stock of machinery, buildings, and equipment. The marginal contribution of labor to GDP therefore diminishes as more of it is added, holding capital constant, exactly as the aggregate production function's diminishing marginal product property predicts.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Correctly identifies that GDP increases by a shrinking amount for each equal increment of labor, explicitly names this as diminishing marginal product, and explains the mechanism (fixed capital shared among more labor).
12-19	12-19	Correctly identifies the diminishing pattern in the data but doesn't explicitly connect it to diminishing marginal product or explain the underlying mechanism.
5-11	5-11	Notes GDP is still rising without identifying that the rate of increase is shrinking.
0-4	0-4	No meaningful attempt, or a fundamental misreading of the table.

A1 (ii)**30 marks**

(ii) How would the aggregate production function of this economy look if GDP is measured along the vertical axis and efficiency units of labor on the horizontal axis?

FULL MODEL ANSWER

The production function would be an upward-sloping curve that is **concave** — it rises throughout its length (more efficiency units of labor always means more GDP), but it rises at a *decreasing rate*, becoming progressively flatter as H increases. This is exactly the shape traced out by the Lithasia data: equal horizontal steps of +4,000 in H produce shrinking vertical steps in GDP (+50, then +30, then +20, then +10).

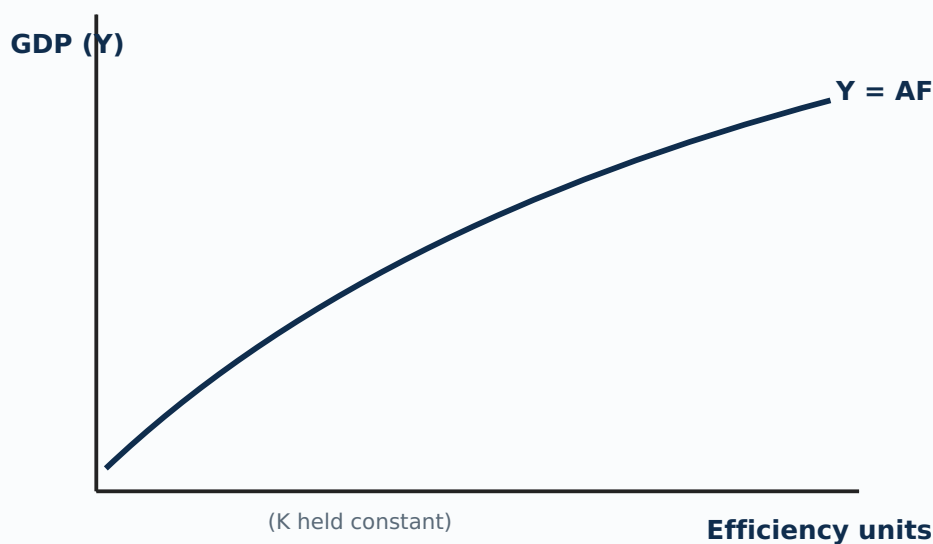


Figure 1: The aggregate production function with efficiency units of labor (H) on the horizontal axis and physical capital (K) held constant. The curve rises throughout but becomes flatter (its slope — the marginal product of labor — steadily decreases) as H increases, reflecting the pattern in the Lithasia data.

Why concave, not a straight line or convex curve. A straight line would imply a constant marginal product of labor (each extra efficiency unit of labor adds the same amount to GDP regardless of how much labor is already employed) — not what the data shows. A convex (increasingly steep) curve would imply an *increasing* marginal product, the opposite of the pattern observed. The concave shape is the graphical signature of diminishing marginal product: the slope of the curve at any point is the marginal product of labor, and that slope visibly gets shallower moving left to right.

This matches Exhibit 6.8 in the standard aggregate production function treatment. Holding physical capital stock constant and plotting GDP against efficiency units of labor produces precisely this concave curve — the mirror-image companion to the (also concave) curve obtained when capital is varied and labor is instead held fixed.

How the 30 Marks Are Likely Allocated

Tier	Marks	What separates this tier
24-30	24-30	Correctly describes/draws a concave, upward-sloping curve, explicitly contrasts it with a straight line (constant marginal product) and a convex curve (increasing marginal product), and connects the shape directly back to the shrinking increments observed in part (i).
15-23	15-23	Correctly identifies the curve as concave/rising at a decreasing rate but without contrasting it against the alternative (straight-line or convex) shapes.
6-14	6-14	Identifies the curve as upward-sloping but doesn't capture the concavity/flattening property at all.
0-5	0-5	No meaningful attempt, or an incorrectly shaped curve (e.g. downward-sloping).

A1 (iii)**45 marks****(iii)** What explains the shape of this aggregate production function?**FULL MODEL ANSWER**

The Law of Diminishing Marginal Product of labor. The aggregate production function has two defining properties: “more is better” (increasing either physical capital or efficiency units of labor, holding the other constant, raises GDP) and the Law of Diminishing Marginal Product (the marginal contribution of either factor to GDP diminishes as more of that factor is used, holding the other factor constant). The concave shape in part (ii) is the direct graphical expression of the second property applied to labor.

The intuitive mechanism: a fixed amount of capital has to be shared among more and more labor. With physical capital held fixed at 15,000 units throughout the Lithasia data, each additional efficiency unit of labor added to the economy has progressively less capital — machinery, buildings, tools, equipment — to work with on average. A firm's workforce doubling while its factory floor space and machine count stay fixed doesn't double output; congestion, the fixed number of machines available at any moment, and the fixed physical workspace all place a growing constraint on how much each additional worker can add. This capital-to-labor ratio falling is the concrete, physical reason each successive block of labor contributes less than the one before.

This mirrors the same principle at the level of an individual firm. The aggregate production function is explicitly modelled on the production function of an individual firm, which exhibits the identical two properties. Just as a single factory adding workers to a fixed number of production lines eventually sees each new hire add less to output than the last, the whole economy adding efficiency units of labor to a fixed aggregate capital stock shows the same diminishing returns, scaled up to the level of GDP.

What would change the shape: an increase in the capital stock itself, or in technology. If physical capital in Lithasia were instead increased alongside labor, GDP could rise faster than the diminishing curve alone would suggest, since more capital gives each unit of labor more to work with again. Similarly, an improvement in technology (the index A in the aggregate production function $Y = AF(K,H)$) would shift the entire curve upward — producing more GDP at every level of H — without changing its fundamentally concave shape, since diminishing marginal product to labor (holding capital fixed) is a property of the function $F()$ itself, not of the technology level A multiplying it.

How the 45 Marks Are Likely Allocated

Tier	Marks	What separates this tier
36-45	36-45	Correctly names the Law of Diminishing Marginal Product as the underlying principle, explains the physical mechanism (a fixed capital stock spread across more labor), connects it to the same property at the level of an individual firm, and discusses what would/wouldn't change the curve's shape (capital increases and technology shifts vs. the shape itself).
22-35	22-35	Correctly names diminishing marginal product and explains the general mechanism, but without the fuller discussion of capital/technology effects on the curve.
9-21	9-21	Recognises returns are diminishing but doesn't correctly explain why (e.g. vague references to "efficiency" without the capital-per-worker mechanism).
0-8	0-8	No meaningful attempt, or a fundamental misunderstanding of the production function's properties.

Question A1 Total: 100 Marks (20% of the Paper)

Question A2

Institutions & Prosperity (100 Marks)

Question A2

100 marks

Look at the following map of Nogales, a twin city that is divided by the U.S. border. One part of Nogales lies in the United States, in Arizona, and the other part lies in Sonora, Mexico. Life in Nogales, Mexico is very different from life in Nogales, Arizona. The average income in Nogales, Mexico is about one-third the average income in Nogales, Arizona. Education levels, life expectancy, and health conditions are better in Nogales, Arizona than in Nogales, Mexico. Unlike Nogales in Arizona, Nogales in Mexico has only recently adopted political reforms, bringing it closer to functioning as a democracy. Crime rates are also lower in Nogales, Arizona than in Nogales, Mexico. Since both cities are located so close to each other, they share similar geographical conditions and climate. The inhabitants of both cities also share a common ancestry and enjoy the same types of food and music.

Required: Based on this information and your own research, what factors do you think can explain why Nogales, Arizona is so much more prosperous than Nogales, Mexico?

FULL MODEL ANSWER

The Nogales case is a textbook natural experiment: geography, climate, ancestry, culture, food, and music are all held constant across the border — the two cities sit within sight of each other. What differs is entirely on the other side of the ledger, and that is precisely the point the example is designed to make. The explanation lies almost entirely in **institutions**, not in geography or culture.

Defining Institutions

Douglass North's definition. Economic historian Douglass North defined institutions as “the rules of the game in a society, or, more formally, the humanly devised constraints that shape human interaction.” Institutions include a country's legal system, property rights protections, electoral and political systems, regulatory bodies, and the informal norms (trust, corruption tolerance, rule-following) that govern how people and firms actually behave.

Why Institutions, Not Geography or Culture, Explain the Gap

The geography/culture explanations are directly ruled out by the setup of the question itself. Nogales, Arizona and Nogales, Mexico share the same climate, terrain, natural resources, ancestry, food, and music — the very factors that geographic or cultural theories of development would point to. Since these are identical on both sides of the border, they cannot be what's driving a threefold income gap between two halves of literally the same city. This is exactly the kind of comparison Daron Acemoglu and James Robinson use in *Why Nations Fail* (2012) to argue that institutions, not geography or culture, are the fundamental cause of prosperity differences — the Nogales case is one of their own central illustrations, alongside the Korean peninsula (see below).

Property rights and the rule of law. In Nogales, Arizona, the U.S. legal system provides strong, reliably enforced property rights and contract enforcement — an entrepreneur can invest, build a business, and trust that courts will protect their ownership and enforce agreements. Where property rights are weak or inconsistently enforced, as historically in much of Mexico, the incentive to invest, start a business, or accumulate capital is undermined, since the returns to that investment are less secure.

Political institutions and democratic accountability. The passage notes Nogales, Mexico “has only recently adopted political reforms, bringing it closer to functioning as a democracy” — a more recently and less deeply embedded democratic system typically comes with weaker checks on the executive, a shorter track record of peaceful transitions of power, and institutions still in the process of building credibility. Acemoglu and Robinson's framework distinguishes **inclusive** institutions (which broadly distribute political power and economic opportunity, incentivising investment and innovation) from **extractive** institutions (which concentrate power and resources among a narrow elite) — the U.S. side has a much longer history of inclusive political institutions than the Mexican side.

Crime, corruption, and the rule of law's enforcement in practice. Lower crime rates in Nogales, Arizona reflect (and reinforce) stronger law enforcement institutions and judicial capacity. High crime and, historically, drug-cartel-related violence on the Mexican side of the border directly deter investment, raise the cost of doing business (security costs, insurance, disrupted supply chains), and discourage entrepreneurship — a direct institutional channel through which weak rule of law translates into lower prosperity.

Access to U.S. markets, capital, and regulatory institutions. Firms and residents in Nogales, Arizona automatically operate within the U.S. federal legal, financial, and regulatory system — deep and liquid capital markets, federally insured banking, and federal infrastructure investment — none of which residents on the Mexican side have equal access to, despite being only metres away. Even identical entrepreneurial talent and effort produce very different outcomes depending on which side of the institutional boundary it's deployed on.

A Parallel Case Strengthens the Argument

North and South Korea. The clearest parallel supporting the institutions hypothesis is the Korean peninsula: a single people, language, culture, and (until 1948) shared history and geography, split into two dramatically different economic outcomes by nothing other than the institutions each half subsequently adopted — South Korea's market-oriented, increasingly inclusive institutions versus North Korea's centrally planned, extractive regime. As with Nogales, every plausible geographic or cultural explanation is held constant, leaving institutional divergence as the explanation that actually varies alongside the outcome.

A strong answer explicitly uses the logic of the comparison (what's held constant vs. what differs) rather than just listing institutional factors in the abstract — the power of the Nogales example is precisely that it isolates institutions as the variable that changes while geography and culture do not.

How the 100 Marks Are Likely Allocated

Tier	Marks	What separates this tier
80-100	80-100	Correctly identifies institutions (property rights, political institutions/democracy, rule of law, crime/corruption, access to capital and regulatory systems) as the explanation, explicitly uses the comparison logic (geography/culture held constant, institutions vary), cites the Acemoglu/Robinson inclusive-vs-extractive framework, and draws the North/South Korea parallel or an equivalent supporting case.
50-79	50-79	Correctly identifies institutions as the key driver with several specific mechanisms, but without explicitly using the comparison logic or citing the relevant literature.
20-49	20-49	Vague reference to "government" or "policy" differences without specific institutional mechanisms, or an answer that leans on geography/culture despite the question's setup ruling these out.
0-19	0-19	No meaningful attempt, or an answer entirely based on factors the question explicitly states are held constant between the two cities.

Question A3

The Population Bomb (100 Marks)

Question A3

100 marks

In his 1968 book *The Population Bomb*, Stanford University professor Paul Ehrlich predicted that rapid population growth would lead to widespread famine and starvation in the 1970s and 1980s, resulting in millions of deaths. However, these predictions did not come true. What were the key flaws in Ehrlich's argument? In your response, explain the factors that helped prevent the mass starvation he predicted.

FULL MODEL ANSWER

Ehrlich's prediction was, at its core, a modern restatement of the **Malthusian** view of population and food supply — and it failed for essentially the same reason the Malthusian cycle itself eventually broke down in the real world: it assumed technology and fertility behaviour would stay fixed while population grew, when in fact both responded to the very pressures Ehrlich was worried about.

The Key Flaw: A Static, Malthusian View of Food Supply

Thomas Malthus's original argument, and its implicit assumption.

Writing in 1798, Thomas Malthus argued humankind was destined to live at the subsistence level: any rise in income above subsistence would trigger higher fertility, population growth would follow, and income per capita would be driven back down to subsistence — a self-defeating cycle with little prospect of sustained improvement. This held reasonably well in pre-modern times specifically because technology advanced too slowly to outrun population growth.

Ehrlich's error was assuming this static relationship would continue to hold indefinitely.

Ehrlich extrapolated 1960s population growth rates forward against a food supply he treated as essentially fixed, or growing only slowly — precisely the Malthusian assumption that population growth outpaces our ability to produce more food. He failed to account for the same two forces that had already allowed Britain, and then other countries, to break out of the Malthusian cycle roughly two centuries earlier: technological progress and the demographic transition.

Factor 1: Technological Progress — The Green Revolution

What actually happened, starting just as Ehrlich was writing. Agronomist Norman Borlaug and colleagues developed high-yield, disease-resistant varieties of wheat and rice through the 1950s and 60s, paired with expanded irrigation, synthetic fertiliser use, and modern farming techniques — a set of innovations that came to be known as the Green Revolution. Global cereal production rose dramatically in the very decades (the 1970s and 1980s) Ehrlich predicted mass famine, with countries like India and Mexico moving from food importers toward self-sufficiency. Borlaug was awarded the Nobel Peace Prize in 1970 specifically for this contribution, which the Nobel committee credited with saving hundreds of millions of lives from starvation.

This is the same mechanism, in miniature, that broke the historical Malthusian cycle. Just as the Industrial Revolution's technological advances allowed sustained income growth to finally outrun population growth after millennia of subsistence-level stagnation, the Green Revolution's agricultural technology allowed food production to outrun the population growth Ehrlich was extrapolating — technology, not a fixed ceiling on food output, determines the real constraint.

Factor 2: The Demographic Transition — Fertility Rates Actually Fell

Ehrlich assumed population growth rates observed in the 1960s would simply continue. In practice, as incomes rose and countries urbanised — the movement of people from rural agriculture into urban manufacturing and services — fertility rates declined substantially across much of the developing world, a pattern known as the demographic transition. This runs directly counter to Malthus's own assumption that rising income fuels higher fertility: historically, once a population moves off the land and income rises past a certain point, fertility tends to fall, not rise, as children shift from being economically productive labour to a costly, education-intensive investment.

Factor 3: Markets and Prices Responded to Scarcity, Rather Than Scarcity Simply Winning

The Simon-Ehrlich wager. Economist Julian Simon publicly challenged Ehrlich's pessimism, betting him in 1980 that the inflation-adjusted prices of five metals (chosen by Ehrlich) would fall, not rise, over the following decade, despite continued population growth — Simon's argument being that rising scarcity itself creates market incentives (higher prices) that drive substitution, more efficient use, exploration, and innovation, ultimately increasing effective supply rather than simply running out. By 1990, all five metals had fallen in real price, and Ehrlich paid out the bet. The episode is now a well-known illustration of the broader point: markets and human ingenuity respond endogenously to scarcity, rather than scarcity simply overwhelming a fixed supply as the raw Malthusian model assumes.

International trade also allowed food to move to where it was needed. Beyond production itself, expanding global trade in agricultural commodities let food surpluses in productive regions reach food-deficit regions, smoothing out what would otherwise have been more severe localised shortages — a redistribution channel Ehrlich's largely closed-economy, fixed-supply framing didn't fully account for.

A strong answer explicitly frames Ehrlich's error as a Malthusian one and shows why: Malthus's cycle only holds if technology and fertility behaviour are static, and the historical record (the Industrial Revolution breaking the original Malthusian cycle, and the Green Revolution plus demographic transition undermining Ehrlich's specific 1968 prediction) shows both are not static — they respond directly to the pressures a growing population creates.

How the 100 Marks Are Likely Allocated

Tier	Marks	What separates this tier
80-100	80-100	Identifies the core flaw as a static, Malthusian assumption about food supply and fertility, explains BOTH the Green Revolution (technology) and the demographic transition (fertility decline) in depth with specific facts (Borlaug, Nobel Prize; urbanisation reducing fertility), and references the Simon-Ehrlich wager or trade as a further supporting mechanism.
50-79	50-79	Correctly identifies the Green Revolution and/or demographic transition as the key factors but with less depth or specific detail, or without framing the error explicitly in Malthusian terms.
20-49	20-49	Vague reference to "technology improved" or "predictions were wrong" without specific mechanisms or facts.
0-19	0-19	No meaningful attempt, or no engagement with why the prediction specifically failed.

Section B — Long Questions

Answer 1 of 2 · 60% of total marks · each part worth 25 marks

Question B1

GDP: Measurement & Its Discontents (100 Marks)

B1 (a)

25 marks

(a) Use the circular flow diagram to show how expenditure, production, and income relate to one another.

FULL MODEL ANSWER

The circular flow diagram shows the economy as two connected loops between households and firms, mediated by two types of markets.

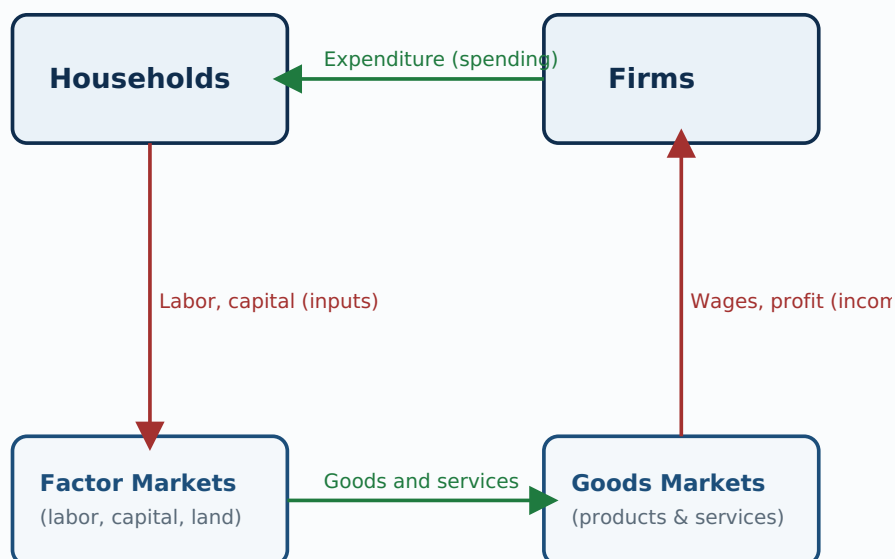


Figure 1: The circular flow diagram. Households supply labor and capital to firms through factor markets, receiving wages and profit (income) in return. Firms supply goods and services to households through goods markets, receiving expenditure (spending) in return. Production, expenditure, and income are three views of the same underlying flow.

The Two Loops

The inner loop — factor markets (red arrows). Households own the economy's factors of production (their own labor, and indirectly, capital through savings and share ownership) and supply them to firms through factor markets. In exchange, firms pay wages, rent, and profit — this is **income** flowing back to households.

The outer loop — goods markets (green arrows). Firms use those factors to **produce** goods and services, which they sell to households through goods markets. Households pay for these goods and services out of their income — this payment is **expenditure**.

Why Production, Expenditure, and Income Must Be Equal

They are three different vantage points on the identical flow of value around the circle. Every euro a firm receives in expenditure from selling its output is, by definition, also the value of what it produced (production), and every euro of that revenue must ultimately be paid out to someone (wages to workers, profit to owners) as income — there is no fourth destination for the money. This is why national income accounting identifies GDP as Production = Expenditure = Income: they are not three independent quantities that happen to be similar, but three equivalent ways of measuring the single flow of value circulating around the diagram.

In practice, GDP is measured using all three approaches as a cross-check. Statistical agencies calculate GDP via the production approach (summing value added across industries), the expenditure approach (summing consumption, investment, government spending, and net exports: $C + I + G + X - M$), and the income approach (summing wages, rents, interest, and profits) — in principle all three yield the same total, and discrepancies between them are used as a check on data quality and to estimate a statistical error term.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Correctly draws/describes both loops (factor markets and goods markets) with households and firms as the two agents, and explicitly explains why production, expenditure, and income must be equal (three views of the same flow), ideally referencing the three GDP measurement approaches.
12-19	12-19	Correctly describes the circular flow but with a less complete explanation of why the three measures are necessarily equal.
5-11	5-11	Partial or unclear diagram/description; households and firms identified but the flows not clearly connected to production/expenditure/income.
0-4	0-4	No meaningful attempt.

B1 (b)**25 marks**

(b) Explain why GDP is considered an inadequate measure of societal progress. Discuss the strengths, as well as the weaknesses of GDP as a measure of societal progress in your answer.

FULL MODEL ANSWER

Strengths of GDP

Objectivity, consistency, and comparability. GDP is calculated using a standardised international methodology (the UN System of National Accounts), letting economies be compared consistently across countries and over time. It is drawn from hard transaction data (sales, wages, tax records) rather than subjective survey responses, giving it a reasonably objective, auditable foundation.

A genuinely useful proxy for material living standards and productive capacity. GDP correlates strongly with many things people plausibly value — access to healthcare, nutrition, education, infrastructure — and it directly measures an economy's capacity to produce goods and services, which remains foundational to a huge range of policy questions (tax revenue capacity, debt sustainability, employment capacity).

Timeliness and regularity. GDP is estimated quarterly (with rapid preliminary estimates) in most developed economies, giving policymakers a regularly updated, comparatively fast signal of the economy's state — something genuinely alternative welfare measures often cannot match in terms of frequency or speed.

Weaknesses of GDP as a Measure of Societal Progress

GDP omits large categories of activity and value entirely. It excludes depreciation of the physical capital stock and natural resources, home production (cooking, cleaning, childcare performed within a household rather than purchased), transactions in the underground or informal economy, and leisure time — despite leisure being something people clearly value and would, in many cases, trade income for.

GDP does not net off negative externalities. Pollution, noise, and crime are not subtracted from GDP — indeed, cleaning up an oil spill or rebuilding after a natural disaster *adds* to GDP (as production of clean-up and reconstruction services), even though society is clearly worse off, not better off, from the underlying event.

GDP says nothing about distribution. A rising GDP is consistent with rapidly rising inequality, where most of the gains accrue to a small share of the population while typical living standards stagnate — the aggregate figure gives no information about how broadly (or narrowly) its growth is shared.

GDP doesn't directly measure well-being or happiness. As the Easterlin Paradox illustrates (discussed further in part (d)), income and subjective wellbeing diverge once basic needs are met — GDP, as a pure production measure, was never designed to capture happiness, health, security, or other dimensions of a good life directly.

A strong answer treats this as a genuine two-sided discussion rather than simply listing GDP's flaws — GDP remains the standard measure precisely because its strengths (objectivity, comparability, direct relevance to production-based policy questions) are real, even as its weaknesses as a welfare or progress measure are also real and well documented.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Discusses genuine strengths (objectivity, comparability, timeliness) AND multiple genuine weaknesses (omitted categories, externalities, distribution, well-being) with specific examples for each, reaching a balanced overall assessment.
12-19	12-19	Covers several weaknesses well but with limited or no discussion of GDP's genuine strengths, producing a one-sided answer.
5-11	5-11	General, vague criticism of GDP without specific mechanisms or examples.
0-4	0-4	No meaningful attempt.

B1 (c)**25 marks**

(c) What alternative metrics to GDP does Diane Coyle suggest in Chapter 4 'Whats it all about' of her book *The Soulful Science*, and how do they aim to provide a more comprehensive understanding of well-being?

FULL MODEL ANSWER

In the middle chapters of *The Soulful Science*, Coyle turns her attention inward to examine several assumptions economists commonly make — including the assumption that GDP per capita is an adequate stand-in for well-being. She surveys several prominent alternatives that have been proposed to capture well-being more directly.

The Alternative Metrics Discussed

The Human Development Index (HDI). Developed by the UN Development Programme, the HDI combines income per capita with measures of health (life expectancy) and education (years of schooling), on the reasoning that a country's development should be judged on more than income alone — a country with high income but poor health or education outcomes should not automatically be ranked as more developed than one with lower income but stronger health and education.

The Index of Sustainable Economic Welfare (ISEW) and its successor, the Genuine Progress Indicator (GPI). Both start from GDP and then adjust it by subtracting a range of social and environmental costs — resource depletion, pollution, and the costs of crime, commuting, and unemployment — and adding the value of non-market activity such as unpaid household labour. By construction, GPI/ISEW growth is typically lower than headline GDP growth, since it's netting off costs that GDP simply ignores.

Subjective wellbeing/happiness surveys. Coyle also discusses the alternative of directly asking people how satisfied or happy they are, rather than inferring welfare from income at all — the approach underlying the Easterlin Paradox research discussed in part (d).

How These Aim to Give a More Comprehensive Picture

Each targets a specific blind spot in GDP. The HDI directly addresses GDP's silence on health and education outcomes; the GPI/ISEW directly addresses GDP's failure to net off environmental and social costs, and its exclusion of unpaid household production; and subjective wellbeing surveys address GDP's complete silence on how people actually experience their lives, rather than what they produce or consume.

Coyle's Own, More Skeptical Conclusion

She does not endorse simply replacing GDP with any single alternative.

Coyle's assessment is notably more cautious than a simple "GDP bad, alternatives good" argument: each alternative index embeds its own, inherently subjective choices about which components matter and how heavily to weight them — the HDI's own weighting of income, health, and education against each other is itself a value judgement, not an objective fact, and different alternative indices can produce quite different rankings of the same countries because of these differing subjective choices. Her broader conclusion leans toward the idea that GDP remains a reasonably good, objective measure of economic *activity* specifically, and that a "dashboard" of several complementary indicators side by side may be more useful and more honest than trying to force everything into one single replacement headline number.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Correctly identifies at least two of the specific alternatives Coyle discusses (HDI, ISEW/GPI, subjective wellbeing surveys) with accurate detail on their construction, explains what blind spot in GDP each addresses, and captures Coyle's own more skeptical/dashboard-favouring conclusion rather than presenting the alternatives as straightforwardly superior to GDP.
12-19	12-19	Correctly names one or two alternatives with reasonable detail but without capturing Coyle's own nuanced conclusion about their limitations.
5-11	5-11	Vague reference to "alternative measures" without naming specific indices or their construction.
0-4	0-4	No meaningful attempt, or no reference to the assigned reading at all.

B1 (d)**25 marks**

(d) How does the "Easterlin Paradox" challenge the assumption that economic growth always correlates with greater happiness? In your answer, discuss potential explanations for the paradox, and its policy implications.

FULL MODEL ANSWER

Economist Richard Easterlin (1974) used national surveys of subjective wellbeing to directly test the assumption, implicit in most economic policy, that rising income and economic growth straightforwardly deliver greater happiness.

The Paradox Itself**Happiness does not rise with wealth once basic needs are met.**

Easterlin's central finding was that within a given country at a point in time, richer individuals do tend to report being happier than poorer individuals — but over time, as a whole country's average income rises substantially, average reported happiness does not rise correspondingly. Lord Layard captured this directly: people in the West have, over recent decades, become considerably richer, work less, travel more, and live longer and healthier lives — “but they are not more happy.” German panel data spanning 1985 to 2000 for the same group of individuals shows real income rising substantially over that period with no corresponding rise in average reported happiness.

The paradox: happiness scales with relative income within a country, but not with absolute income growth over time. This is genuinely puzzling under a simple model where more income mechanically buys more of what people want. If richer people within a country are reliably happier than poorer people in the same country, why doesn't an entire country becoming richer over time make everyone happier together?

Potential Explanations

Relative income and social comparison. One leading explanation is that happiness depends less on absolute income than on income *relative* to others around you. If everyone's income rises together, relative position (and therefore the social comparison that seems to drive much of reported happiness) stays roughly the same, even though everyone is objectively better off in absolute terms.

Hedonic adaptation. People appear to adapt psychologically to improved material circumstances fairly quickly — a higher income or a bigger house delivers a temporary boost to happiness that fades as it becomes the new normal, a phenomenon sometimes called the “hedonic treadmill.” Under this explanation, sustained happiness gains from income growth are inherently difficult to achieve since the reference point simply moves up alongside income.

Non-income needs become more binding once basic needs are met.

Once food, shelter, and safety are secured, additional happiness may depend far more on factors GDP doesn't capture at all — social relationships, work-life balance, autonomy, health, and community — rather than further income gains, which is consistent with the observation that the income-happiness link is much stronger among poorer individuals and countries than among already-wealthy ones.

Policy Implications

Growth-maximisation may not be the right single policy objective. If the paradox holds, a government purely focused on maximising GDP growth may be optimising for a measure that has already largely stopped tracking what it presumably cares about (citizen wellbeing) once a country reaches a reasonably developed income level — strengthening the case (echoing part (c)) for policymakers to track broader wellbeing indicators alongside GDP, not merely GDP growth alone.

Policies targeting relative position, security, and non-income factors may matter more than further growth. If relative income and social comparison genuinely drive much of reported happiness, policies addressing inequality directly, strengthening job and health security, protecting leisure time, and supporting community and social connection may raise wellbeing more effectively, per euro of policy effort, than policies purely aimed at accelerating GDP growth for already-developed economies.

A caveat: the paradox's robustness has been genuinely contested. Later research (notably by Betsey Stevenson and Justin Wolfers) using longer and more comparable cross-country datasets has challenged whether the original paradox holds as strongly as Easterlin found, suggesting richer countries may show somewhat higher average happiness than poorer ones after all, even if the relationship is far from one-to-one — a balanced answer notes the paradox remains genuinely debated in the literature, not a fully settled empirical fact.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Correctly explains the paradox (happiness rises with income within a country at a point in time, but not with income growth over time), offers at least two genuine explanations (relative income/social comparison, hedonic adaptation, non-income needs) and at least one specific, well-reasoned policy implication, ideally noting the paradox's contested empirical status.
12-19	12-19	Correctly explains the paradox and offers one explanation and one implication, with less depth.
5-11	5-11	Vague reference to "money doesn't buy happiness" without correctly capturing the specific within-country vs. over-time distinction Easterlin identified.
0-4	0-4	No meaningful attempt.

Question B1 Total: 100 Marks (60% of the Paper)

Question B2

Business Cycles, Multipliers & Stabilisation Policy (100 Marks)

B2 (a)**25 marks**

(a) What are the major sources of fluctuations identified by each of these three schools of thought (i) real business cycle theory (ii) Keynesian theory, and (iii) financial and monetary theories?

FULL MODEL ANSWER

Macroeconomists have identified three broad schools of thought explaining what actually drives short-run economic fluctuations, each emphasising a different underlying source of shocks.

(i) Real Business Cycle Theory

Source of fluctuations: productivity and technology shocks. Real business cycle theory attributes fluctuations to changes in productivity and technology. A wave of productivity-enhancing innovation causes an economic expansion, as firms can produce more output from the same inputs. Conversely, a rise in the price of a key input — oil being the classic example — acts as a negative supply shock and can trigger a recession, since it raises production costs economy-wide independent of demand conditions.

(ii) Keynesian Theory

Source of fluctuations: shifting expectations and "animal spirits." Keynesian theory focuses on changes in business and consumer expectations about the future. "Animal spirits" — John Maynard Keynes's term for the psychological factors driving shifts in business and consumer mood or sentiment — can independently trigger either decreases in spending (recessions) or increases in spending (expansions), regardless of whether any change in underlying productivity or fundamentals has actually occurred. A negative shock can generate pessimism; willingness to spend falls, and — critically for the Keynesian mechanism — that initial fall in spending is not offset by increased spending elsewhere in the economy, and is instead amplified further through multiplier effects as one person's reduced spending becomes another person's reduced income.

(iii) Financial and Monetary Theory

Source of fluctuations: changes in the money supply, prices, and interest rates. Associated principally with Milton Friedman, this view attributes fluctuations to monetary factors: a decrease in the money supply (M2) causes the price level to fall, which reduces employment because wages are downwardly rigid (they don't easily adjust down to match falling prices, so real labour costs effectively rise, and firms cut jobs). A falling money supply also raises the real interest rate, which reduces investment spending by firms, compounding the contractionary effect.

A strong answer keeps the three theories genuinely distinct rather than blending them — RBC theory locates the shock in the real, productive side of the economy (technology/productivity); Keynesian theory locates it in psychology and expectations (demand-side, amplified by multipliers); and financial/monetary theory locates it in the money supply and interest rates. Real-world recessions are often explained using elements of more than one theory simultaneously, which is a reasonable point to make, but the theories themselves are analytically separate.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Correctly identifies the distinct source of fluctuations for all three schools (RBC: productivity/technology shocks; Keynesian: animal spirits/expectations amplified by multipliers; financial/monetary: money supply, price level, and interest rates), with the specific mechanism explained for each rather than just named.
12-19	12-19	Correctly identifies two of the three theories with reasonable mechanism detail, the third vague or missing.
5-11	5-11	Names the three schools but without correctly distinguishing their underlying mechanisms.
0-4	0-4	No meaningful attempt.

B2 (b)**25 marks**

(b) The concept of multipliers was one of the key elements of John Maynard Keynes's theory of fluctuations. What is a multiplier? Explain with an example.

FULL MODEL ANSWER

A **multiplier** is the mechanism by which an initial change in spending generates a larger total change in economy-wide income and output, because one person's spending becomes another person's income, which is then partly re-spent, generating further rounds of income and spending down the chain.

The Mechanism

Spending circulates and compounds, rather than stopping after one transaction. When one euro of new spending occurs, the recipient of that euro (a worker, a supplier, a landlord) doesn't save it entirely — they spend a portion of it themselves, which becomes income for someone else, who in turn spends a portion of that, and so on. The total increase in economy-wide income ends up larger than the initial injection of spending, with the exact size of the multiplier depending on how much of each round is re-spent versus saved or spent on imports (which leak out of the domestic spending chain).

Multipliers work in both directions, and are central to Keynesian theory precisely because they amplify shocks. An initial fall in spending (driven, say, by a negative shift in animal spirits) doesn't just directly reduce the income of whoever was going to receive that spending — it reduces their spending too, which reduces someone else's income, and so on, amplifying the original shock into a much larger overall contraction. This amplification is exactly why Keynesian theory treats multipliers as one of the central elements explaining how economic fluctuations can become so much larger than the shocks that triggered them.

A Worked Example: The Páirc Uí Chaoimh Redevelopment

The initial injection. Consider the redevelopment of Páirc Uí Chaoimh, the GAA stadium in Cork. The construction spend itself is the initial injection — wages paid to construction workers, payments to building material suppliers, and fees paid to architects and contractors.

The first round of re-spending. The construction workers who receive those wages don't save all of it — they spend a good share of it locally: in Cork city cafes, shops, and pubs near the site, on transport, and on other everyday goods and services. That spending becomes income for the owners and staff of those local businesses.

Further rounds, and the eventual total effect. Those café and shop owners, in turn, spend part of their increased income on their own suppliers, staff wages, and purchases — and so the chain continues, each round smaller than the last as some income is saved or spent outside the local area (a “leakage”) rather than re-spent locally. The eventual total increase in economic activity generated across Cork city, once every round of this chain is added up, is larger than the construction spend alone — this total effect (original spend plus all the induced rounds of re-spending) is the multiplier effect, and is exactly the kind of local economic impact argument regularly used to justify major stadium and infrastructure redevelopments.

A complete answer explains *why* the chain eventually stops (leakages to saving, taxation, and imports mean each round is progressively smaller, so the total effect converges to a finite multiple of the initial spend rather than growing without limit) rather than simply asserting spending “multiplies” without explaining the mechanism or its limit.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Correctly defines the multiplier mechanism (spending becomes income, partly re-spent, generating further rounds), explains why the effect eventually diminishes and converges (leakages to saving/imports), and applies it in a fully worked, specific example (such as Páirc Uí Chaoimh) tracing at least two rounds of the chain.
12-19	12-19	Correctly defines the multiplier concept with a relevant example, but without tracing multiple rounds or explaining why the effect eventually diminishes.
5-11	5-11	Vague reference to spending having “knock-on effects” without a clear mechanism or worked example.
0-4	0-4	No meaningful attempt.

B2 (c)**25 marks**

(c) What do countercyclical policies mean? During an economic boom, what countercyclical fiscal and monetary policies are used? Please use a diagram of the business cycle to help explain your answer.

FULL MODEL ANSWER

Countercyclical policies are fiscal and monetary policy actions deliberately taken to move *against* the current direction of the business cycle — restraining the economy during a boom (when it is running above trend, as in Figure 2) and stimulating it during a recession (when it is running below trend) — in order to smooth out fluctuations and keep actual GDP closer to its trend path, rather than reinforcing whichever direction it happens to be moving in already.

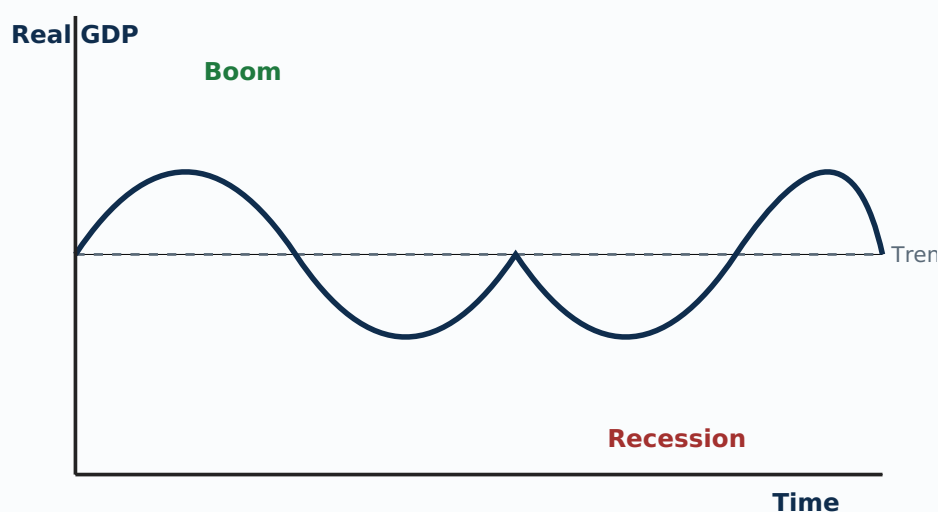


Figure 2: The business cycle — actual real GDP (solid line) fluctuating around its long-run trend (dashed line). A boom is a period where GDP rises above trend; a recession is a period where it falls below trend.

Countercyclical Policy During a Boom

The logic: cool down an economy that is overheating. During a boom, GDP is running above its sustainable trend, which risks fuelling inflation, asset bubbles, and unsustainable borrowing. Countercyclical policy during a boom aims to withdraw some stimulus from the economy specifically to prevent these excesses, rather than letting the boom run unchecked toward a sharper eventual correction.

Countercyclical Fiscal Policy in a Boom

Reduce government spending and/or raise taxes. A government pursuing countercyclical fiscal policy during a boom would restrain its own spending growth and/or raise taxes, withdrawing some private demand from an already-strong economy — the opposite of the spending increases/tax cuts used to fight a recession. In practice this also means resisting the temptation to spend windfall tax revenues (which rise automatically during a boom as incomes and profits rise) and instead saving them, for instance into a sovereign wealth or rainy-day fund, for use during the next downturn.

Countercyclical Monetary Policy in a Boom

Raise interest rates. A central bank pursuing countercyclical monetary policy during a boom raises interest rates, which increases the cost of borrowing for both households and firms, cooling consumption and investment spending, and helping contain inflationary pressure building up from the strong economy — the reverse of the interest rate cuts (and, in more extreme downturns, quantitative easing) used to stimulate a recession.

A common error is describing expansionary policies (spending increases, tax cuts, rate cuts) when asked specifically about boom-time countercyclical policy — those are the correct countercyclical response to a *recession*, precisely the opposite direction to what's being asked here. Countercyclical, by definition, means pushing against the current phase of the cycle: contractionary policy in a boom, expansionary policy in a recession.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Correctly defines countercyclical policy as moving against the current cycle phase, correctly identifies contractionary fiscal policy (spending restraint/tax increases) AND contractionary monetary policy (interest rate rises) as the boom-time response, and uses a correctly labelled business cycle diagram distinguishing boom from recession relative to trend.
12-19	12-19	Correctly identifies contractionary fiscal and monetary policy for a boom but without a clear diagram, or with one of the two policy types missing.
5-11	5-11	Describes expansionary (recession-fighting) policies instead of the boom-time contractionary policies actually asked for.
0-4	0-4	No meaningful attempt.

B2 (d)**25 marks**

(d) Critically evaluate the fiscal and monetary policies of the Irish government since the turn of the 21st century.

FULL MODEL ANSWER

Since Ireland adopted the euro in 1999, its own government has had no independent monetary policy lever at all — interest rates are set for the whole eurozone by the European Central Bank (ECB). This answer therefore evaluates Irish fiscal policy (which Ireland fully controls) across four distinct periods, and evaluates monetary policy specifically in terms of how well a single ECB rate has fitted Ireland's own economic conditions at each stage.

2000–2008: The Celtic Tiger Boom — Pro-Cyclical Fiscal Policy

Fiscal policy reinforced the boom rather than restraining it. Rather than pursuing countercyclical policy (as discussed in part (c)) by restraining spending and banking windfall revenue during the boom, Irish fiscal policy through the 2000s was broadly **pro-cyclical**: government spending and tax cuts both expanded significantly, funded substantially by transaction-based taxes (stamp duty, VAT on construction) tied directly to an overheating property market. This left the exchequer's revenue base structurally exposed to a property market correction.

Monetary policy: a single ECB rate that was a poor fit for an overheating Ireland. As part of the eurozone, Ireland faced the same ECB policy rate as much larger, more sluggish economies like Germany and France. Those rates were arguably too low for an Irish economy running well above its sustainable trend — a textbook “one size fits none” problem of monetary union, since the single rate could not simultaneously suit an overheating Ireland and a stagnating German economy.

2008–2013: Financial Crisis, Bank Guarantee, and the Bailout

A blanket bank guarantee and severe fiscal consolidation. Following the property crash and global financial crisis, the government issued a blanket guarantee of the banking sector's liabilities in September 2008, and subsequently pumped tens of billions of euros into recapitalising and nationalising banks (including Anglo Irish Bank), pushing the sovereign's own debt burden to unsustainable levels. Ireland formally entered an €85 billion EU-IMF-ECB ("Troika") assistance programme in December 2010, exiting in December 2013 having implemented severe fiscal consolidation (spending cuts and tax rises, including a 14% cut to public sector pay in 2009) as a condition of the programme.

Evaluation: necessary, but arguably compounded a downturn that pro-cyclical boom-time policy had made worse. The scale of austerity required was directly a consequence of the fiscal position built up (and left exposed) during the pro-cyclical boom years — a clear illustration of why countercyclical fiscal discipline during a boom matters, since its absence left far less room to manoeuvre once the downturn hit. On the monetary side, Ireland had no independent lever to cushion the shock at all (no ability to devalue its own currency or set its own interest rate), leaving fiscal austerity as essentially the only adjustment tool available.

2014-2019: Recovery

A strong recovery, aided in part by ECB accommodation. Ireland's economy recovered strongly through this period, with unemployment falling from a peak of over 15% in 2011-2012 to under 6% by 2019. The ECB's own quantitative easing programme (asset purchases) from 2015 kept borrowing costs low across the eurozone, indirectly supporting Ireland's recovery and debt sustainability even though Ireland had no direct control over that policy.

2020-Present: COVID-19, and a Return of Fiscal Windfalls

A large, temporary fiscal expansion during COVID-19. The government ran significant deficits to fund the Pandemic Unemployment Payment, the Employment Wage Subsidy Scheme, and broader health and business supports — a genuinely countercyclical fiscal response to an exceptional shock, in contrast to the pro-cyclical pattern of the 2000s.

Since 2022-2023: a corporate tax windfall, and the return of an old lesson only partially learned. Ireland has recorded historically large budget surpluses since 2022, driven overwhelmingly by concentrated corporate tax receipts from a small number of large multinationals — corporate tax receipts reached €23.8 billion in 2023, and the 2024 surplus reached roughly 7% of modified GNI, boosted further by a €13 billion Apple back-tax payment following an EU Court of Justice ruling. To its credit, the government has explicitly avoided simply spending this windfall pro-cyclically, instead channelling it into two new sovereign wealth vehicles (the Future Ireland Fund and the Infrastructure, Climate and Nature Fund) targeting combined assets of over €100 billion by the mid-2030s — showing at least a partial, hard-won lesson learned from the concentrated, transaction-tax-dependent revenue base that left the exchequer so exposed in the 2000s. That said, the underlying revenue base remains heavily concentrated in a handful of large taxpayers, leaving genuine ongoing volatility risk the government itself openly acknowledges.

Monetary policy since 2022: another rate mismatch, in the opposite direction. The ECB's aggressive rate hikes from 2022 to fight eurozone-wide inflation were arguably a tighter setting than Ireland's own inflation and growth conditions alone would have called for, again illustrating the structural cost of ceding monetary policy entirely to a currency union average.

A strong answer makes the throughline explicit: Ireland's fiscal policy has swung between pro-cyclical (2000s boom, arguably worsening the eventual crash) and genuinely countercyclical (COVID-19 response, and more recent windfall saving via the sovereign wealth funds), while its monetary policy has been entirely determined by the ECB throughout, for better (crisis-era low rates, QE) and worse (a policy rate that has periodically been too loose or too tight for Ireland's own specific conditions) — the central, recurring critical point being that fiscal policy is the only lever Ireland itself actually controls.

How the 25 Marks Are Likely Allocated

Tier	Marks	What separates this tier
20-25	20-25	Covers multiple distinct periods (Celtic Tiger boom, financial crisis/bailout, recovery, COVID-19, recent surplus) with specific facts for each, explicitly evaluates fiscal policy as pro-cyclical vs. countercyclical at different points, correctly notes Ireland's monetary policy is entirely determined by the ECB, and reaches a genuinely critical (not merely descriptive) overall assessment.
12-19	12-19	Covers two or three periods with reasonable factual detail but with limited critical evaluation, or doesn't clearly address the ECB/monetary policy dimension.
5-11	5-11	General, largely undated description of "austerity" or "recession" without specific periods, figures, or critical evaluation.
0-4	0-4	No meaningful attempt.

Question B2 Total: 100 Marks (60% of the Paper)